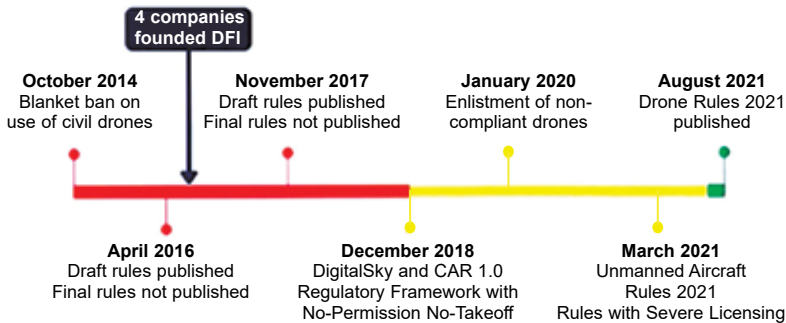


Understanding drone regulations in India

Navigating the skies with drones has become a popular endeavour worldwide. In India, this is no exception. However, as with every form of technology that occupies public space, there are guidelines and regulations to ensure safety and security. Decoding the zones:



Red zone. These zones represent the strictest areas for drone operations, often covering sensitive locations like military bases, defence areas, or other restricted areas. Just as one wouldn't approach the Indian Prime Minister uninvited, flying drones in these zones without permission is forbidden. They are marked red to signal a 'no-fly zone.'

Yellow zone. Zones near airports or areas with frequent aircraft operations pose dangers from drones to larger aircraft. Consider a drone obstructing a commercial jet's path at Heathrow; the outcome could be disastrous. In these zones, drones need explicit permissions. A yellow zone is akin to visiting an office: not forbidden, but one must verify their intent and credentials before entry.

Green zone. Here, the skies are most open to drone enthusiasts. Consider these as public roads in the sky, where drone pilots can operate freely, given that they respect general guidelines. Essentially, it's akin to the free movement on a public road.

To determine zone classifications, refer to the government's Digital Sky platform, which is similar to Google Maps. While India encourages drone innovation, operators must understand these zones. Certifications and permissions aim to integrate drones safely into India's airspace.

dent that the current provisions are just the tip of the iceberg. The complex ecosystem of drone operations, from hardware and software to data transmission and storage, presents many entry points for potential cyber threats. However, the silver lining is the government's proactive approach. Even at this nascent stage of drone integration into Indian skies, there's a dedicated focus on software and hardware tamper-proofing in the certification scheme. Moreover, ongoing discussions and revisions indicate a commitment to upgrade and strengthen these cybersecurity measures continuously. Collaboration will be key. Engaging with global certification bodies, standardisation organisations, and cybersecurity experts will ensure

India's drones are technologically advanced and fortified against cyber threats.

While the future of drones in India promises to transform various sectors, this transformation must be underpinned by robust cybersecurity measures. As the Indian drone ecosystem continues to evolve, it's evident that while we might not have all the answers yet, we are certainly asking the right questions.

The future is unmanned

Unmanned aerial vehicles, or drones, are more than just the future; they're the present. They represent a paradigm shift in viewing transportation, warfare, surveillance, and many other applications. The industry is booming, with over 400

companies in India now involved in drone component manufacturing and assembly. The potential is evident, with drones expected to play a pivotal role in shaping the future of various sectors.

The drone industry is thriving in India. The data is shown in Fig. 3.

Drones offer a unique employment opportunity in India. Traditional skills often require months of training, but drones only need a 10th-grade education, basic English proficiency, and five-day training. This training can lead to jobs with handsome salaries. Many drone pilots earn even more, with some reaching salaries comparable to airlines pilots. The drone industry is rapidly growing, with over 400 companies now involved in component manufacturing and assembly, especially in agriculture, healthcare, and survey mapping. This growth, especially noticeable in the past year, showcases drones' potential for job creation.

It's an exciting time to be part of the drone ecosystem in India, whether as a manufacturer, investor, or enthusiast. As the world moves towards more autonomous solutions, drones will undoubtedly be at the helm, leading the way. So, here's to soaring high, with eyes in the sky, and embracing the future drones promise. The drone revolution in India is well underway. As these unmanned aerial vehicles continue to make inroads into various sectors, they're proving to be more than just technological marvels. They're tools of transformation, bringing efficiency, safety, and innovation. With the government's support and industry growth, the sky's the limit for drones in India. **EFY**

This article is from a tech talk session at EFY Expo Delhi 2023 by Smit Shah, President Drone Federation of India. Transcribed and curated by Akanksha Sondhi Gaur, Research Analyst and Journalist at EFY.